

Auditory setup

McGinley lab

The setup shown in the McGinley lab's slides, with a microphone kit added.

Item	Count	Source	Cost (\$)
PXIe-1082 equipment chassis	1	eBay / used	2,650.00
PXIe-6363 signal input/output board	1	eBay / used	2,495.36
PXI-4461 audio module, hybrid 186900-11L	1	eBay / used	699.99
PXIe-8360 chassis connection module	1	eBay / used	493.20
PCIe-8361 computer connection card	1	eBay / listed new	197.00
779500-03 rack cable, 3 metres	1	eBay / listed new	89.85
SHC68-68-EPM cable, 192061C-02	1	eBay / used	89.99
BNC-2090A signal connection panel	1	eBay / used	613.77
ED1 driver + enclosure/power supply	1	eBay / used	209.06
ES1 speaker + matching cable	1	Estimate	500.00
Hosa BNC-58-103 output lead	1	Pro Audio / new	13.53
Added microphone and calibration kit	1	See Compact	2,517.22
Estimated parts total			10,568.97

SONIC

This follows the sound setup used in the SONIC paper. We could consider this setup.

Item	Count	Source	Cost (\$)
USB-6361 sound generation/recording, 782255-01	1	Newark / new	3,744.42
TimeMachines TM2500C time reference	1	New / backorder	799.99
MF1-M speaker kit	1	Estimate	1,500.00
SA1 speaker amplifier	1	Estimate	2,000.00
ZB1PS powered enclosure	1	Estimate	750.00
781513-01 power supply	1	Newark / new	123.42
780534-01 locking computer cable	1	DigiKey / new	69.75
763830-01 United States power cord	1	Newark / new	51.00
781233-02 cable noise filter	1	DigiKey / new	38.25
Hosa BNC-58-103 output lead	1	Pro Audio / new	13.53
Added microphone and calibration kit	1	See Compact	2,517.22
Estimated parts total			11,607.58

Basic

For tones below 20 kilohertz.

Item	Count	Source	Cost (\$)
Behringer UMC204HD sound interface	1	eBay / new	97.98
JBL 305P MkII powered speaker	1	Sweetwater / new	99.00
Dayton EMM-6 measurement microphone	1	eBay / new	66.93
Hosa HMIC-010 microphone cable	1	Sweetwater / new	17.99
Hosa HSS-005 audio cable, \$14.95 each	2	Sweetwater / new	29.90
Gator RI-MICTP-FBM microphone stand	1	Sweetwater / new	34.99
REED R8090 sound-level calibrator	1	DigiKey / new	219.00
Parts total			565.79

Compact

A simpler setup for the paper's full frequency range.

Item	Count	Source	Cost (\$)
USB-6361 sound generation/recording, 781442-01	1	Newark / new	3,030.42
MF2-M speaker kit	1	Estimate	1,500.00
SA1 speaker amplifier	1	Estimate	2,000.00
ZB1PS powered enclosure	1	Estimate	750.00
Pomona EM4970-36# output lead	1	DigiKey / new	23.68
Hosa PHX-106F-BULK microphone connection lead	1	B&H / new	13.95
Yageo 100,000-ohm resistor, \$0.10 each	2	DigiKey / new	0.20
763830-01 United States power cord	1	Newark / new	51.00
<i>Microphone and calibration kit</i>			
Earthworks M50 measurement microphone	1	eBay / new	1,299.00
Sound Devices MM-1 microphone amplifier	1	B&H / new	945.00
REED R8090 sound-level calibrator	1	DigiKey / new	219.00
Hosa HMIC-010 microphone cable	1	Sweetwater / new	17.99
Gator RI-MICTP-FBM microphone stand	1	Sweetwater / new	34.99
Panasonic LR6XWA/2SB battery, \$0.62 each	2	DigiKey / new	1.24
Estimated parts total			9,886.47